

Failure Mode and Effects Analysis (FMEA)

Document ID:	FMEA-B2M-001
Product:	NEXUS B2M (30420)
Version:	1.5
Effective Date:	2024-06-01
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1. Purpose

This FMEA evaluates potential failure modes in the manufacturing process of NEXUS B2M (Reference 30420) Beta-2 Microglobulin immunoassay kits. The analysis highlights assembly and changeover risks following shared production line usage with other NEXUS products.

2. Scope

Applicable to the complete NEXUS B2M manufacturing process at Marcy-l'Etoile. Shared assembly line AL-003 is used for multiple NEXUS products including NEXUS CK MB and NEXUS Troponin.

3. FMEA Risk Assessment

Process Step	Failure Mode	Effect	S	Cause	O	Controls	D	RPN
Kit_Assembly	Incomplete line clearance	Wrong component in kit	9	Rushed changeover	2	Clearance checklist, operator training	3	54
Kit_Assembly	Wrong SPR cone type	Cross-contamination between products	9	Short changeover time	3	Visual verification, barcode scan	4	108
SPR_Receipt	SPR coating degradation	Reduced assay sensitivity	7	Storage temperature excursion	2	Temperature monitoring, incoming QC	3	42
Strip_Assembly	Incorrect reagent volume	Assay inaccuracy	7	Dispenser calibration drift	2	Daily dispenser verification	3	42
Calibration	Calibrator reconstitution error	Invalid calibration curve	8	Operator technique variation	2	Training programme, visual aids	4	64

Process Step	Failure Mode	Effect	S	Cause	O	Controls	D	RPN
QC_Testing	Contaminated control sample	False QC failure	5	Improper sample handling	2	Aseptic technique training, single-use aliquots	3	30
Labeling	Wrong product label applied	Product mix-up at customer site	9	Label roll changeover error	1	Camera-based label verification system	2	18
Packaging	Missing insert / IFU	Customer unable to perform test	5	Packaging station error	2	Weight verification, checklist	3	30

4. Risk Priority Summary

Critical threshold: RPN >= 80. One failure mode exceeds the critical threshold:

- **Wrong SPR cone type (RPN=108):** This is the highest-priority risk. Assembly line changeover procedure SOP-ASM-003 mandates a minimum 30-minute changeover time with an 8-item clearance checklist. Barcode scanning of SPR cone sachets is required before kit assembly begins.

Moderate risks (RPN 50-80):

- **Incomplete line clearance (RPN=54):** Mitigated by operator training and clearance checklist per SOP-ASM-003.
- **Calibrator reconstitution error (RPN=64):** Visual aids and standardised training programme in place.

5. Revision History

Version	Date	Author	Changes
1.0	2023-01-10	L. Renard	Initial release
1.5	2024-06-01	L. Renard	Added wrong SPR cone type risk (RPN=108); updated changeover controls